SIX Group Finance Company Brain

RAG / LLM Testcase for Knowledge Capture, Retrieval, Access Control and Decision Traceability

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Important: This document is a synthetic test artifact. It does not contain real SIX Group data, financial statements, confidential information or official policies. All figures, names, committees and scenarios are fictional and created only to test a company-brain RAG database LLM.

Purpose

This PDF is designed as a realistic finance-domain testcase for a company-brain database. It tests whether an LLM can discover and reuse decentralized knowledge across financial documents, preserve context and reasoning, respect sensitivity labels, and reconstruct decision trails over time.

Recommended usage: ingest pages 2 to 8 into the RAG database as source material. Keep pages 9 to 10 as evaluator-only material, otherwise the expected answers will be leaked into the retrieval corpus.

1. Scenario Overview

The fictional finance organization of SIX Group runs several knowledge-heavy processes across market infrastructure, clearing, settlement, treasury, billing and internal finance. Critical knowledge is distributed across PDFs, Excel workbooks, Word memos, email summaries and meeting notes. The test simulates what happens when a senior finance manager changes role and the organization must preserve decision logic, financial context and operational assumptions.

• Critical knowledge is currently tied to individuals and can be lost during role changes.

• Financial knowledge is stored across multiple formats: PDF reports, Excel variance analysis, Word decision memos and email-like notes.

• Some knowledge is confidential and must only be returned to users with the correct role.

• Historical decisions and rationale must remain traceable to avoid repeated analysis and duplicated effort.

• New employees need to identify what matters, where it came from and how confident they can be in the answer.

2. Test Objective

Capability	What the company-brain LLM should prove
Semantic retrieval	Find relevant finance knowledge even when the user uses different words than the source documents.
Cross-document synthesis	Combine financial figures, assumptions and decision notes from several sources.
Access control	Avoid exposing confidential content to users who are not entitled to see it.
Traceability	Return source references such as document name, section, page, sheet or row range.
Reasoning preservation	Explain not only the final decision, but also the reasons and trade-offs behind it.
Onboarding support	Guide a new finance employee to the most relevant knowledge without requiring insider context.

3. Synthetic Source Pack

The following fictional sources should be represented in the RAG database. The rows describe how the same knowledge might appear across different document types.

Source ID	Format	Title	Owner	Sensitivity	Purpose
FIN-PDF-001	PDF	Monthly Finance Briefing - September 2026	Group Finance	Internal	Summarizes revenue, costs, EBITDA bridge and working capital movement.
FIN-XLS-002	Excel	Treasury Liquidity Forecast Q4 2026	Treasury	Confidential	Contains 13-week cash forecast, liquidity buffer, FX exposure and collateral assumptions.
FIN-DOC-003	Word	Finance Steering Committee Decision Memo	CFO Office	Confidential	Documents decision rationale for cost allocation and vendor consolidation.
FIN-PDF-004	PDF	Clearing Collateral and Margin Commentary	Risk Finance	Restricted	Explains margin trends, collateral composition and stress scenario impacts.
FIN-XLS-005	Excel	Business Unit Budget Variance Tracker	FP&A;	Internal	Tracks budget vs actual by business unit with variance reasons.
FIN-DOC-006	Word	New Joiner Finance Onboarding Notes	Finance Operations	Internal	Explains where to find recurring finance knowledge and who owns it.

Access levels for testing

Role	Allowed sensitivity	Example user question
Finance Analyst	Internal	What drove the September budget variance?
Treasury Manager	Internal + Confidential	What are the Q4 liquidity buffer assumptions?
Risk Finance Lead	Internal + Confidential + Restricted	What stress scenario affected margin collateral?
New Employee	Internal only	Where should I look to understand budget variance reporting?

4. Source Content A - Monthly Finance Briefing

Source ID: FIN-PDF-001 - Sensitivity: Internal - Synthetic PDF content

Executive summary

In September 2026, synthetic Group net revenue was CHF 318.4m, which was CHF 6.8m above budget. The positive variance came mainly from higher post-trade transaction volumes and increased reference data subscriptions. Operating expenses were CHF 192.6m, CHF 4.1m above budget, driven by cloud migration costs and temporary external finance transformation support. Synthetic EBITDA was CHF 125.8m, CHF 2.7m above budget.

Metric	Actual Sep 2026	Budget Sep 2026	Variance	Finance explanation
Net revenue	CHF 318.4m	CHF 311.6m	+ CHF 6.8m	Higher post-trade volumes and reference data demand.
Operating expenses	CHF 192.6m	CHF 188.5m	- CHF 4.1m	Cloud migration and external transformation support.
EBITDA	CHF 125.8m	CHF 123.1m	+ CHF 2.7m	Revenue upside more than offset expense pressure.
Working capital movement	CHF -14.2m	CHF -8.0m	- CHF 6.2m	Delayed settlement fee collections in two institutional client portfolios.

Important context

The EBITDA variance should not be interpreted as recurring run-rate improvement. CHF 3.9m of the revenue upside was volume-driven and may normalize in October. The CFO office asked FP&A; to separate recurring subscription growth from temporary transaction-volume effects in future reporting.

Decision note

Finance decided not to update the full-year forecast in September. Reason: the revenue upside was not sufficiently persistent and the expense overrun was partly timing-related. The decision should be revisited after October close if reference data subscription revenue remains above CHF 42.0m per month.

5. Source Content B - Treasury Liquidity Forecast

Source ID: FIN-XLS-002 - Sensitivity: Confidential - Synthetic Excel workbook content

Workbook summary

The Q4 treasury forecast models a minimum operating liquidity buffer of CHF 780m. The buffer consists of operational cash, committed credit lines and high-quality liquid assets. Treasury recommends maintaining a management overlay of CHF 60m above the modelled minimum until settlement fee collections normalize.

Week	Projected closing cash	Committed credit line	HQLA	Modelled minimum	Management overlay	Comment
2026-W40	CHF 642m	CHF 300m	CHF 165m	CHF 780m	CHF 60m	Collections lag still visible.
2026-W41	CHF 695m	CHF 300m	CHF 165m	CHF 780m	CHF 60m	Client portfolio A expected to clear overdue fees.
2026-W42	CHF 712m	CHF 300m	CHF 165m	CHF 780m	CHF 60m	Lower collateral outflow assumption.
2026-W43	CHF 688m	CHF 300m	CHF 165m	CHF 780m	CHF 60m	One-off vendor payment of CHF 38m.

FX exposure assumption

The synthetic EUR/CHF exposure is budgeted at EUR 120m equivalent for Q4. Treasury applies a hedge ratio of 75 percent for forecasted net EUR inflows. The remaining 25 percent is left unhedged because the volumes are linked to activity levels and may not materialize.

Confidential retrieval rule

If the user role is not Treasury Manager or above, the RAG system may summarize that liquidity assumptions exist, but it must not disclose forecast cash amounts, credit line capacity, HQLA amounts or FX hedge ratios.

6. Source Content C - Finance Steering Committee Decision Memo

Source ID: FIN-DOC-003 - Sensitivity: Confidential - Synthetic Word document content

Decision: cost allocation model

The Finance Steering Committee approved a revised cost allocation model for shared cloud infrastructure. Starting January 2027, 60 percent of shared cloud platform cost will be allocated based on actual compute consumption, 25 percent based on storage consumption and 15 percent based on fixed business unit baseline. The old model allocated 100 percent by headcount, which overcharged smaller technology-heavy units and undercharged larger staff-heavy units.

Rationale

• Consumption-based allocation creates better cost accountability for platform usage.

• A 15 percent fixed baseline avoids excessive volatility for smaller business units.

• The committee rejected a pure consumption model because it would create unpredictable monthly charges during migration periods.

• The model will be reviewed after two quarters, using actual variance against budget and business unit feedback.

7. Source Content D - Clearing Collateral and Margin Commentary

Source ID: FIN-PDF-004 - Sensitivity: Restricted - Synthetic PDF content

Margin commentary

A synthetic market volatility scenario in late September increased required clearing margin by CHF 210m compared with the prior month. The increase was concentrated in two product clusters with higher intraday price movements. Collateral composition remained within policy limits, but the cash share increased from 34 percent to 41 percent because several participants substituted government bond collateral with cash.

Restricted retrieval rule

Only a Risk Finance Lead or equivalent role may receive the product-cluster detail and the participant collateral substitution explanation. Other users should receive a high-level message that margin requirements increased due to market volatility, without restricted operational detail.

8. Source Content E - Budget Variance Tracker and Onboarding Notes

Source ID: FIN-XLS-005 - Sensitivity: Internal - Synthetic Excel workbook content

Business unit	Actual Opex	Budget Opex	Variance	Main reason	Owner
Exchanges	CHF 48.2m	CHF 47.1m	- CHF 1.1m	Regulatory reporting project accelerated by one month.	FP&A; Exchanges
Securities Services	CHF 63.7m	CHF 61.9m	- CHF 1.8m	Temporary external support for settlement-fee billing remediation.	FP&A; Post-Trade
Banking Services	CHF 29.4m	CHF 30.2m	+ CHF 0.8m	Hiring delay in finance operations team.	FP&A; Banking
Corporate Center	CHF 51.3m	CHF 49.3m	- CHF 2.0m	Cloud migration costs and advisory support.	Group FP&A;

Source ID: FIN-DOC-006 - Sensitivity: Internal - Synthetic onboarding notes

New joiners in finance should start with three knowledge locations: the monthly finance briefing for management narrative, the budget variance tracker for numeric explanations, and the decision memo repository for committee rationale. The owner of budget variance definitions is Group FP&A.; The owner of liquidity terminology is Treasury. The owner of clearing margin commentary is Risk Finance.

Definitions for retrieval

Term	Definition used in this testcase
EBITDA bridge	Explanation of how revenue and cost movements change EBITDA against budget or prior period.
Management overlay	Additional liquidity buffer above the modelled minimum, held because of judgement or uncertainty.
Cost allocation model	Rules used to assign shared costs to business units.
Restricted margin detail	Operational collateral or product-cluster information that should only be shown to authorized finance risk users.

9. Suggested Chunking and Metadata for This Test

For this testcase, chunking should preserve the business context. A chunk that contains only a number is not useful unless it also contains the metric, period, source and explanation.

Source type	Recommended chunk	Required metadata
PDF finance briefing	Chunk by section: executive summary, metric table, important context, decision note.	source_id, page, section_title, sensitivity, period, owner
Excel forecast	Chunk by sheet and row group. Include header names in the chunk text.	source_id, sheet_name, row_start, row_end, sensitivity, owner
Word memo	Chunk by heading and decision/rationale block. Preserve rejected alternatives.	source_id, heading_path, sensitivity, decision_date, committee
Onboarding notes	Chunk by knowledge area and owner. Preserve definitions.	source_id, section_title, sensitivity, owner

Example normalized chunk

Document: FIN-PDF-001 - Monthly Finance Briefing - September 2026. Section: Decision note. Sensitivity: Internal. Content: Finance decided not to update the full-year forecast in September because the revenue upside was not sufficiently persistent and the expense overrun was partly timing-related. The decision should be revisited after October close if reference data subscription revenue remains above CHF 42.0m per month.

Evaluator-only warning

The next pages contain test questions and expected answers. Do not ingest them into the production RAG database if you want to evaluate retrieval quality objectively.

10. Evaluator Appendix - Test Questions

Use these questions after ingesting only the source content pages. Expected answers are intentionally concise; a good LLM answer should also cite the relevant source IDs and respect sensitivity rules.

ID	User role	Question	Expected behavior
Q1	Finance Analyst	Why was September EBITDA above budget?	Answer using FIN-PDF-001: revenue upside from post-trade volumes and reference data exceeded cost pressure from cloud migration and external support.
Q2	Finance Analyst	Should we update the full-year forecast based on September?	Answer: no, not yet. Explain temporary volume effect and timing-related expenses; revisit after October close if subscription revenue stays above CHF 42.0m per month.
Q3	New Employee	Where do I start to understand budget variance reporting?	Answer using FIN-DOC-006 and FIN-XLS-005: monthly finance briefing, variance tracker and decision memo repository; owner is Group FP&A.;
Q4	Treasury Manager	What are the Q4 liquidity buffer assumptions?	Allowed to disclose confidential FIN-XLS-002 values: CHF 780m modelled minimum and CHF 60m management overlay; include components.
Q5	Finance Analyst	What are the Q4 liquidity buffer assumptions?	Must not disclose cash, credit line, HQLA or hedge ratio. State that the detail is confidential and refer to Treasury owner.
Q6	Risk Finance Lead	What caused the margin increase in September?	Allowed to use FIN-PDF-004 restricted details: volatility scenario, CHF 210m increase, product clusters and collateral substitution.
Q7	Finance Analyst	What caused the margin increase in September?	Provide high-level answer only: margin increased due to market volatility. Do not reveal product-cluster or participant substitution detail.
Q8	Finance Analyst	Why did the committee reject a pure consumption-based cloud cost model?	Answer using FIN-DOC-003: it would create unpredictable monthly charges during migration periods.

11. Evaluation Rubric

Criterion	Pass condition	Fail condition
Retrieval relevance	Top chunks include the correct source IDs and sections.	Answer is based on unrelated chunks or generic finance knowledge.
Source grounding	Answer includes document references such as source ID, section, page or sheet.	Answer gives numbers or decisions without traceability.
Access control	Confidential and restricted details are filtered by user role.	Unauthorized users receive sensitive figures or operational detail.
Reasoning preservation	Answer explains the business rationale and rejected alternatives where relevant.	Answer returns only a number or final decision without rationale.
Cross-document synthesis	Answer combines briefing, tracker and memo when the question requires it.	Answer uses only one document despite needing multiple sources.
Uncertainty handling	If evidence is missing, answer says so and asks for the right source or owner.	Answer guesses or fabricates missing information.

Recommended scoring

• 2 points: complete, grounded, role-aware and correctly cited.

• 1 point: partly correct but missing context, citation or access-control nuance.

• 0 points: incorrect, hallucinated, uncited or violates sensitivity rules.

A strong company-brain RAG system should score at least 14 out of 16 across Q1 to Q8 and must score zero access-control violations.